Wuao Liu

Contact

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Research Interests

Computer vision and machine learning, with a focus on learning visual representations through multimodal supervision. Problems of interest include multimodal learning (audio-visual integration, vision-language tasks), and video understanding. The application domains span AI for science, AI for robotics, etc.

Education

University of Massachusetts Amherst — Aug. 2024 – Apr. 2029 (expected)
Ph.D. in Computer Science — Amherst, MA
Advisor: Dr. Grant Van Horn

University of Michigan — Aug. 2021 – Apr. 2023
M.S. in Robotics — Ann Arbor, MI
Advisor: Dr. Jason Corso

Zhejiang University (ZJU) — Sep. 2017 – Jun. 2021
B.Eng. in Automation — Hangzhou, China

Publications

1. Mustafa Chasmai, **Wuao Liu**, Subhransu Maji, and Grant Van Horn. **Audio Geolocation: An Investigation with Natural Sounds**. *Under Submission.*
2. Filippos Bellos, Yayuan Li, **Wuao Liu**, and Jason Corso. **Can Large Language Models Reason About Goal-Oriented Tasks?** In *Proceedings of ACL Workshop on the Scaling Behavior of Large Language Models*, pages 24-34, 2024. [paper]

Work Experience

The Corso Group, UMich — Jun. 2023 – Apr. 2024
Research Engineer — Ann Arbor, MI
Mentor: Dr. Jason Corso
Project: Perceptually-enabled Task Guidance (PTG). We adopted foundation models (`OpenAI Whisper, GPT models`) for a VR/AR system to assist humans in cooking. My responsibilities included training action recognition models and developing an Automatic Speech Recognition (ASR) module.

Tencent AI Lab — Jun. 2021 – Aug. 2021
Machine Learning Engineer Intern — Shenzhen, China
Mentor: Dr. Peilin Zhao, Dr. Mingyang Sun
Project: We developed a multi-agent reinforcement learning algorithm for optimizing electric vehicle charging and navigation systems using real-world data.

Research Experience

Computer Vision Lab, UMass — Aug. 2024 – Dec. 2024
Advisor: Dr. Grant Van Horn — Amherst, MA
Project: **Contrastive Learning of Audio and Geographical Data**

- Proposed the first work to geolocate natural audio at the global scale.
- Investigated the potential of species sounds as geolocation signals with strong oracles and proof of concepts.
- Enabled multimodal (audio, image, text) geolocation with cross-modal contrastive learning.

UM Ford Center for Autonomous Vehicles May. 2022 – Aug. 2022
Advisor: Dr. Katie Skinner Ann Arbor, MI
Project: **Neural Radiance Field (NeRF) for Autonomous Driving**

- Developed an open-source tool to collect RGB-D images in the CARLA simulator and visualize the corresponding camera poses.
- Implemented a scene-specific NeRF and achieved a PSNR of 28.29 dB for novel view synthesis in urban scenarios.
- Gained hands-on experience with multi-camera calibration using the Kalibr toolbox and worked with various types of cameras, such as event, RGB-D, and thermal cameras.

Presentations

Audio Geolocation: An Investigation with Natural Sounds, New England Computer Vision (NECV) Workshop, Yale University. (November 2024)

Technical Skills

- Programming Languages: Python, C/C++, MATLAB, LaTeX.
- Deep Learning: PyTorch, TensorFlow, Hugging Face, Scikit-learn.
- Robotics Frameworks: ROS, GTSAM.